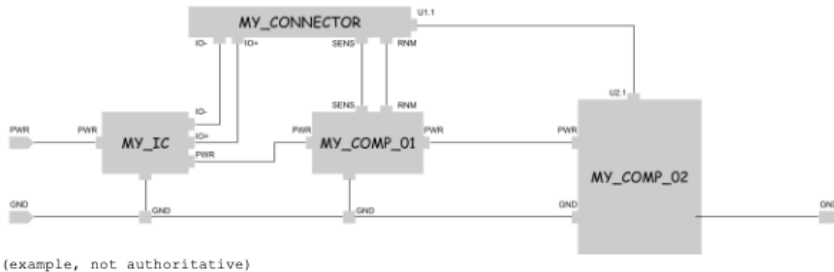


Quilter Backend Engineer Code Challenge

PCB Board Renderer



(example, not authoritative)

Parse the ECAD JSON files and generate a visual representation of a PCB layout.

Task

1. Board Boundary

- Draw the board outline from ``boundary.coordinates``
- This forms the outer edge of the PCB

2. Components

- Draw component outlines at their transformed positions
- Show component reference designators (R1, C1, U1)
- Handle rotation correctly

3. Traces

- Draw traces as lines/paths with their specified ``width``
- Use different colors for different layers

4. Vias

- Draw vias as circles at their ``center`` positions
- Diameter from ``diameter`` field

5. Keepout Regions

- Draw keepout areas with a distinct pattern or color
- Mark them as restricted areas

6. Return an error if the board cannot be parsed.

- Return a message of the failure. 14 of the boards will fail to parse.

Requirements

- Render a board as a SVG, PNG, or PDF
- Board renders correctly and is easy readable
- Solution can be reviewed and verified within 20 minutes
- Includes unit testing and proper code commenting

Usage of LLMs is acceptable